Kai Wu, J. Andrew Zhang, and Y. Jay Guo

nd Accurate Linear Fitting for an Incompletely ed Gaussian Function With a Long Tail

eriment data onto a curve is a signal processing technique data features and establish hip between variables. Of- ct the curve to comply with cal function and then turn nto estimating the unknown f a function. Among analyti- for data fitting, the Gaussian e most widely used one due ve applications in numerous engineering fields. To name Gaussian function is highly tatistical signal processing thanks to the central limit and the Gaussian function ppears in the quantum har- ator, quantum field theory, s, and many other theories n physics [2]; moreover, the nction is widely applied in r depicting molecular orbit- ter science for imaging pro- in artificial intelligence for al networks.

Gaussian function, or, sim- n fitting, is consistently of t to the signal processing [3]–[6]. Since the Gaussian nderlain by an exponential s nonlinear and not easy to ectly. One effective way of g its exponential nature is to tural logarithm, which has been applied in transferring the Gaussian fitting into a linear fitting [4]. However, the problem of the logarithmic transformation is that it makes the noise power vary over data samples, which can result in biased Gaussian fitting. The weighted least square (WLS) fitting is known to be effective in handling uneven noise backgrounds [7]. However, as unveiled in [5], the ideal weighting for linear Gaussian fitting is directly related to the unknown Gaussian function. To this end, an iterative WLS is developed in [5], starting with using the data samples (which are noisy values of a Gaussian function) as weights and then iteratively reconstructing the weights using the previously estimated function parameters.

For the iterative WLS, the number of iterations required for a satisfactory fitting performance can be large, particularly when an incompletely sampled Gaussian function with a long tail is given [see Figure 1(a) for such a case]. Establishing a good initialization is a common strategy for improving the convergence speed and performance of an iterative algorithm. Noticing the unavailability of a proper initialization for the iterative WLS, we aim to fill the blank by developing a high-quality one in this article. To do so, we introduce a few signal processing tricks to develop high-performance initial estimators of the iterative WLS substantially improved, but its accuracy is also greatly enhanced, particularly for noisy and incompletely sampled Gaussian functions with long tails. These will be demonstrated by simulation results.

Prior art and motivation

Let us start by elaborating on the signal model. A Gaussian function can be written as

$$f(x) = Ae^{-\frac{(x-\mu)^2}{2\sigma^2}}, \tag{1}$$

where x is the function variable, and A, μ, and σ are the parameters to be estimated. They represent the height, location, and width of the function, respectively. Directly fitting $f(x)$ can be cumbersome due to the exponential function. A well-known opponent of the exponential is the natural logarithm. Indeed, by taking the natural logarithm of both sides of (1), we can obtain the following polynomial after some basic rearrangements:

$$\ln(f(x)) = a + bx + cx^2, \tag{2}$$

where the coefficients a, b, and c are related to the Gaussian function parameters μ, σ, and A. Based on (1) and (2), it is easy to obtain

$$\mu = \frac{-b}{2c}; \quad \sigma = \sqrt{\frac{-1}{2c}}; \quad A = e^{a-b^2/(4c)}.$$

a, b, and c are coefficients
mial, they can be readily
y employing linear fitting
g., the least square (LS)

signal processing, we gener-
noisy digital signals. Thus,
) given in (1), the following
e likely to be dealt with:

$$+\xi[n],\quad \text{s.t. } f[n]=f(n\delta_x),$$
$$N-1, \tag{4}$$

he sample index, δ_x is the
rval of x, and $\xi[n]$ is an ad-
Gaussian noise. If we take the
ithm of $y[n]$, we then have

$$\begin{aligned}
&\ln(f[n]+\xi[n])\\
&\ln\left(f[n]\left(1+\frac{\xi[n]}{f[n]}\right)\right),\\
&\ln(f[n])+\frac{\xi[n]}{f[n]}\\
&a+b\delta_x n+c\delta_x^2 n^2+\frac{\xi[n]}{f[n]},
\end{aligned} \tag{5}$$

where the first-order Taylor series $\ln(1+x)\approx x$ is applied to get the approximation, and $\ln(f[n])$ is written into a polynomial form based on (2). Next, we review several linear fitting methods through which the motivation of this work will be highlighted.

For the sake of conciseness, we employ vector/matrix forms for the sequential illustrations. In particular, let us define the following three vectors:

$$\begin{aligned}
\mathbf{y}&=[\ln(y[0]),\ln(y[1]),\ldots,\\
&\quad\ln(y[N-1])]^{\mathrm{T}},\\
\mathbf{x}&=[0,\delta_x,\ldots,(N-1)\delta_x]^{\mathrm{T}},\\
\boldsymbol{\theta}&=[a,b,c]^{\mathrm{T}}.
\end{aligned} \tag{6}$$

Then, based on (5), the linear Gaussian fitting problem can be conveniently written as

$$\mathbf{y}=\mathbf{X}\boldsymbol{\theta}+\boldsymbol{\xi},\quad \text{s.t.}\quad \mathbf{X}=[\mathbf{1},\mathbf{x},\mathbf{x}\odot\mathbf{x}], \tag{7}$$

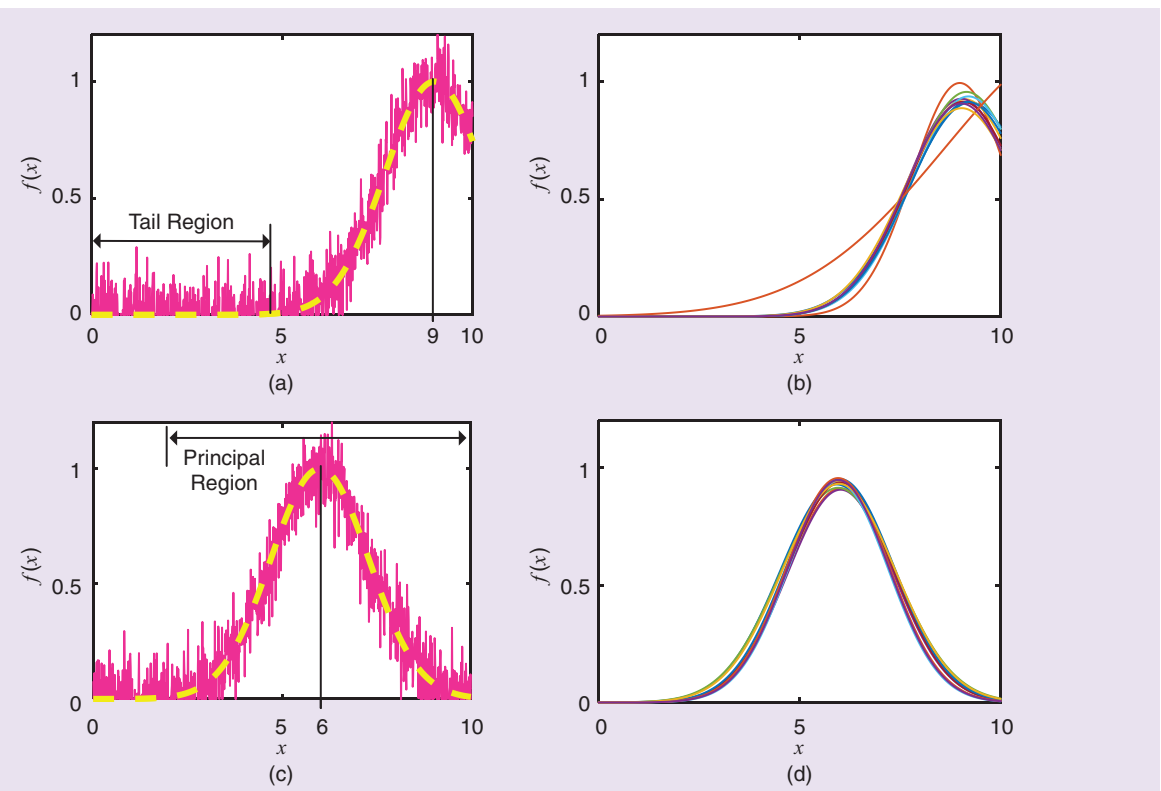
where $\boldsymbol{\xi}$ denotes a column vector stacking the noise terms $\xi[n]/f[n]$ $(n=0,1,\ldots,N-1)$ in (5), and $\odot$ denotes the pointwise product. Due to the use of these vector/matrix forms, the estimators reviewed in the following sections look different from their descriptions in the original work. However, regardless of the forms, they are the same in essence.

LS fitting

The first fitting method [4] reviewed here applies LS on (7) to estimate the three unknown coefficients in $\boldsymbol{\theta}$. The solution is classical and can be written as [7]

$$\hat{\boldsymbol{\theta}}=\mathbf{X}^{\dagger}\mathbf{y}=(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}}\mathbf{y}, \tag{8}$$

where $\mathbf{X}^{\dagger}$ denotes the pseudoinverse of $\mathbf{X}$. The LS fitting is simple but not without problems. As can be seen from (5), the additive white Gaussian noise $\xi[n]$ is divided by $f[n]$. The division can severely increase the noise power at ns with $f[n]\approx 0$, causing the noise enhancement problem. As a consequence of the problem, LS can suffer from poor



rmance, particularly when
of samples are from the tail
Gaussian function, such as
ed in Figure 1(a).

S fitting

noise enhancement issue,
of [5] propose replacing LS
In contrast to LS, which
ample equally, WLS applies
ghts over samples. The pur-
hting is to counterbalance
ons. For this purpose, $f[n]$
weight, as seen from (5).
$n]$ is the digital sample of
Gaussian function.
the problem, an iterative
eloped in [5], where $y[n]$
rsion of $f[n]$) is used as the
t. From the second itera-
ght is constructed using the
mates of a, b, and c based
on depicted in (2). Let $\mathbf{w}_i$
veighting vector at the ith
llecting the weights over
$N-1$. Then, the iterative
executed as

$$\mathbf{X}_i^{\mathrm{T}}\mathbf{X}_i)^{-1}\mathbf{X}_i^{\mathrm{T}}\mathbf{y}_i, \quad i=0,1,\ldots$$
$$_i \odot \mathbf{y}; \quad \mathbf{X}_i=[\mathbf{w}_i, \mathbf{w}_i \odot \mathbf{x},$$
$$_i \odot \mathbf{x} \odot \mathbf{x}]$$
$$\text{if } i=0; \quad \text{otherwise,}$$
$$_i = e^{\mathbf{x}\hat{\boldsymbol{\theta}}_{i-1}}, \tag{9}$$

$\mathbf{y}$ are given in (6) and $\mathbf{X}$ in
onential is calculated point-
ast row.
a sufficiently large number
iterative WLS can achieve a
ance Gaussian fitting, gen-
than LS. However, the more
he more time-consuming
S would be. Moreover, the
mber of iterations to achieve
ting performance changes
portion of the tail region.
Gaussian function in Fig-
an illustration. The tail re-
t half the whole sampled
orm 12 numbers of itera-
on (9) for 10 trials, each
pendently generated noise
that some fitting results still substantially differ from the true function, even after 12 iterations. In contrast, perform the same fitting as described but change μ to six (equivalently reducing the tail region). The fitting results of 10 independent trials are plotted in Figure 1(d). Obviously, the results look much better than those in Figure 1(b).

A common solution to reducing the number of iterations required by an iterative algorithm is a good initialization. In our case, the quality of the initial weight vector, i.e., $\mathbf{w}_0$ in (9), can affect the overall number of iterations required by the iterative WLS to converge. Moreover, if the way of initializing $\mathbf{w}_0$ can be immune to the proportion of the tail region, the convergence performance of the iterative WLS can then be less dependent on the proportion of the tail region. Our work is mainly aimed at designing a way of initializing the weight vector of the iterative WLS so as to reduce the number of overall iterations and relieve the dependence of fitting performance on the proportion of the tail region.

An interesting but not-good-enough initialization

An initialization for iterative WLS-based linear Gaussian fitting is developed in [6], which was originally motivated by separately fitting the parameters of a Gaussian function. In particular, exploiting the following relation

$$\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = A\sqrt{2\pi}\,\sigma, \tag{10}$$

the work proposes a simple estimation of σ, as given by

$$\hat{\sigma} = \frac{1}{\hat{A}}\sum_{n=0}^{N-1} y[n]\,\delta_x, \tag{11}$$

where $y[n]$ is the noisy sample of the Gaussian function to be estimated, δ_x is the sampling interval of x, and the summation approximates the integral of $f(x)$ described earlier. Moreover, $\hat{A}$ is estimated by

$$[\hat{A}, \hat{n}] = \max y[n], \tag{12}$$

mization is achieved. According to the middle relation in (3), c can be estimated as $\hat{c} = -1/2\hat{\sigma}^2$. Thus, the work in [6] removes c from the parameter vector $\boldsymbol{\theta}$ given in (6) and employs an iterative WLS, similar to (9) but with reduced dimension, to estimate a and b.

A major error source of $\hat{\sigma}$ obtained in (11) is the approximation error from using the summation in (11) to approximate the integral given in (10). Even if δ_x is fine enough, the approximation can still be problematic, depending on the proportion of the tail region in the sampled Gaussian function. For example, if the sampled function has a shape similar to the one given in Figure 1(c), we know that the summation can well approximate the integral given a fine δ_x. However, if the sampled function has a shape like the curve in Figure 1(a), the approximation error will be large regardless of δ_x. A condition is given in [6] stating when the summation in (11) can well approximate the integral in (10). Nevertheless, the question "What shall we do when the condition is not satisfied?"—which can be inevitable in practice—is yet to be answered.

Proposed Gaussian fitting

Looking at Figure 1(a), we know the summation in (11) cannot approximate the integral in (10). Now, instead of using the summation to approximate something unachievable, how about looking into a different question: "What can be approximated using the summation given in (11)?" We answer this question by performing the following computations (which look complex but are easily understandable):

$$\begin{aligned}
\sum_{n=0}^{N-1} y[n]\,\delta_x &\overset{(a)}{\approx} \sum_{n=0}^{N-1} \delta_x f[n] \overset{(b)}{\approx} \int_0^{N\delta_x} f(x)\,\mathrm{d}x \\
&\overset{(c)}{=} \int_0^{\mu} f(x)\,\mathrm{d}x + \int_{\mu}^{N\delta_x} f(x)\,\mathrm{d}x \\
&\overset{(d)}{=} \frac{1}{2}\int_0^{2\mu} f(x)\,\mathrm{d}x \\
&\quad + \frac{1}{2}\int_{\mu-(N\delta_x-\mu)}^{N\delta_x} f(x)\,\mathrm{d}x \\
&\overset{(e)}{=} \frac{\sqrt{2\pi}A\sigma}{2}\,\mathrm{erf}\!\left(\frac{\mu}{\sigma\sqrt{2}}\right)
\end{aligned}$$

is the so-called error func-
e defined as [1]

$$= \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} \mathrm{d}t. \qquad (14)$$

ep in (13) is obtained is de-
ows:
step replaces $y[n]$ with
omitting the noise term
given in (4).
s how the integral is often
ed in a math textbook.
e left side of $\overset{(b)}{\approx}$ approxi-
alculates the area below
ght side does so exactly.
ntegrating interval is split
guous halves.
ntegrating interval of each
n $\overset{(c)}{=}$ is doubled in such a
it becomes symmetric
$x = \mu$. Since $f(x)$ is also
ic against $x = \mu$, see (1):
ing coefficient $1/2$ can
lance the extension of the
g interval.
ased on a known fact [1]:

$$)\mathrm{d}x = \sqrt{2\pi} A\sigma \,\mathrm{erf}\Big(\frac{\epsilon}{\sigma\sqrt{2}}\Big).$$

$\overset{(c)}{=}$ in (13), we can also split
on $\sum_{n=0}^{N-1} y[n]\delta_x$. Doing so,
grals on the right-hand side
) can be, respectively, ap-
by

$$\sum_{n=0}^{\hat{n}-1} y[n]\delta_x \quad \text{and}$$
$$\sum_{n=\hat{n}}^{N-1} y[n]\delta_x, \qquad (15)$$

obtained in (12). (Note that
stimate of μ.) Moreover,
computations in (13), we
tain

$$\frac{\hat{A}\sigma}{} \,\mathrm{erf}\Big(\frac{\hat{n}\delta_x}{\sigma\sqrt{2}}\Big);$$
$$\frac{\hat{A}\sigma}{} \,\mathrm{erf}\Big(\frac{N\delta_x - \hat{n}\delta_x}{\sigma\sqrt{2}}\Big), \qquad (16)$$

μ have been replaced by
es given in (12). The two

ence of the nonelementary function $\mathrm{erf}(\cdot)$, analytically solving σ from the equations is nontrivial. Moreover, we have two equations but one unknown. How to constructively exploit the information provided by both equations is also a critical problem. Here, we first develop an efficient method to estimate σ from either equation in (16), resulting in two estimates of σ; we then derive an asymptotically optimal combination of the two estimates.

Efficient estimation of σ

The two equations in (16) have the same structure. Therefore, let us focus on the top one for now. While solving σ analytically is difficult, numerical means can be resorted to. As is commonly done, we can select a large region of σ, discretize the region into fine grids, evaluate the values of the right-hand side in (16) on the grids, and identify the grid that leads to the closest result to S_β.

These steps are regular but not practically efficient. This is because evaluating the right-hand side of the equation in (16) needs the calculation of $\mathrm{erf}(\cdot)$ for each σ grid. From (14), we see that $\mathrm{erf}(\cdot)$ itself is an integral result. If we calculate $\mathrm{erf}(\cdot)$ onboard, it would be approximated by a summation over sufficiently fine grids of the integrating variable. This can be highly time-consuming, particular given that $\mathrm{erf}(\cdot)$ needs to be calculated for each entry in a large set of σ grids. Alternatively, we may choose to store a lookup table of $\mathrm{erf}(\cdot)$ onboard. This is doable but can also be troublesome, for the reason that the parameter of $\mathrm{erf}(\cdot)$, as dependent on the parameters of the Gaussian function to be fitted, can span over a large range in different applications. The trouble, however, can be relieved through a simple variable substitution.

Making the substitution of $\hat{n}\delta_x = \mathsf{k}\sigma$ in (16), we obtain

$$\mathsf{S}_\beta \approx \frac{\sqrt{2\pi}\hat{A}\hat{n}\delta_x}{2\mathsf{k}} \,\mathrm{erf}\Big(\frac{\mathsf{k}}{\sqrt{2}}\Big). \qquad (17)$$

Clearly, the dependence of the $\mathrm{erf}(\cdot)$ function on μ (represented by $\hat{n}\delta_x$ and σ, as shown in (16), is now removed, making the $\mathrm{erf}(\cdot)$ function solely related to the coefficient k. Therefore, a significance of the variable substitution is that one lookup table of $\mathrm{erf}(\cdot)$ can be applied to a variety of applications with different Gaussian function parameters. Assuming $k = \mathsf{k}^\star$ makes the right-hand side of (17) closest to S_β, σ can then be estimated as $\hat{n}\delta_x/\mathsf{k}^\star$. Similarly, we can make the substitution for the bottom equation in (16) and obtain another estimate of σ. In summary, the two estimators can be established as in (18) shown at the bottom of the page. The two estimates would have different qualities, depending on how many samples are used for each. This further suggests that combining them is not as trivial as simply averaging them. Next, we develop a constructive way of combining them.

Asymptotically optimal σ estimation

The two estimates obtained in (18) are mainly differentiated by how many samples are used in their estimations. Therefore, identifying the impact of the employed samples on the estimation performance is helpful in determining a way to combine the estimates. One of the most common performance metrics for an estimator is the Cramér–Rao lower bound (CRLB) [7]. This points the direction of our next move.

Let us check the CRLB of $\hat{\sigma}_\beta$ first. It is estimated using the samples $y[n]$ for $n = 0, 1, \ldots, \hat{n}-1$. Referring to (4), $y[0]$, $y[1], \ldots, y[\hat{n}-1]$ are jointly normally distributed with different means

$$\hat{\sigma}_\alpha = \frac{(N-\hat{n})\delta_x}{\mathsf{k}^\star}; \quad \text{s.t.} \quad \mathsf{k}^\star : \mathrm{argmin}_{\mathsf{k}} \Big(\mathsf{S}_\alpha - \frac{\sqrt{2\pi}\hat{A}(N-\hat{n})\delta_x}{2\mathsf{k}} \,\mathrm{erf}\Big(\frac{\mathsf{k}}{\sqrt{2}}\Big)\Big)^2;$$

variance, as given by σ_ξ^2. σ_ξ^2 is the power of the noise $\forall n$) given in (4). Since we n investigating the estima- ance of σ, we assume that known. Then, the CRLB ation, as obtained based on ., $y[\hat{n}-1]$, can be com-

$$= \frac{\sigma_\xi^2}{\sum_{n=0}^{\hat{n}-1}\left(\frac{\partial f[n]}{\partial \sigma}\right)^2} = \frac{\sigma_\xi^2}{\frac{\sum_{n=0}^{\hat{n}-1} f[n]^2(\mu-\delta_x n)^4}{\sigma^6}}, \tag{19}$$

middle result is a simple of [7, Eq. (3.14)], and the derivative of $f[n]$ can be ved based on its expression With reference to (19), we write the CRLB of $\hat{\sigma}_\alpha$, as

$$\} = \frac{\sigma_\xi^2}{\frac{\sum_{\hat{n}}^{N-1} f[n]^2(\mu-\delta_x n)^4}{\sigma^6}}, \tag{20}$$

sole difference com- (19) is the set of ns for ion.

specting the two CRLBs, hey only differ by a linear amely,

$$\frac{\}}{\}} = \frac{\sum_{n=\hat{n}}^{N-1} f[n]^2(\mu-\delta_x n)^4}{\sum_{n=0}^{\hat{n}-1} f[n]^2(\mu-\delta_x n)^4}. \tag{21}$$

rom this result is that we linear combination of the mates obtained in (18) an asymptotically optimal stimation. To further illus- us consider the following nation:

$$(1-\rho)\hat{\sigma}_\beta, \quad \rho \in (0,1), \tag{22}$$

$$\begin{aligned}\mathbb{E}\{(\hat{\sigma}-\sigma)^2\} &= \mathbb{E}\{(\rho\hat{\sigma}_\alpha + (1-\rho)\hat{\sigma}_\beta \\ &\quad - \rho\sigma - (1-\rho)\sigma)^2\} \\ &= \rho^2\mathbb{E}\{(\hat{\sigma}_\alpha-\sigma)^2\} \\ &\quad + (1-\rho)^2\mathbb{E}\{(\hat{\sigma}_\beta-\sigma)^2\} \\ &\sim \rho^2\mathrm{CRLB}\{\hat{\sigma}_\alpha\} \\ &\quad + (1-\rho)^2\mathrm{CRLB}\{\hat{\sigma}_\beta\},\end{aligned} \tag{23}$$

where "$a \sim b$" denotes that a asymptotically approaches b or a constant linear scaling of b.

Solving $\partial\mathbb{E}\{(\hat{\sigma}-\sigma)\}/\partial\rho = 0$ leads to the following optimal ρ:

$$\begin{aligned}\rho^\star &= \frac{\mathrm{CRLB}\{\hat{\sigma}_\beta\}}{\mathrm{CRLB}\{\hat{\sigma}_\beta\} + \mathrm{CRLB}\{\hat{\sigma}_\alpha\}} \\ &= \frac{\sum_{n=\hat{n}}^{N-1} f[n]^2(\mu-\delta_x n)^4}{\sum_{n=0}^{N-1} f[n]^2(\mu-\delta_x n)^4},\end{aligned}$$

where the CRLB expressions (19) and (20) have been applied. The optimality of $\rho^\star$ can be validated by plugging $\rho = \rho^\star$ into (23). Doing so yields

$$\mathbb{E}\{(\hat{\sigma}-\sigma)^2\} \sim \frac{\sigma_\xi^2}{\frac{\sum_{n=0}^{N-1} f[n]^2(\mu-\delta_x n)^4}{\sigma^6}}. \tag{24}$$

From the index ranges of the summations in (19), (20), and (24), we can see that the right-hand side of (24) becomes the CRLB of the σ estimation that is obtained based on all samples at hand—the best estimation performance for any unbiased estimator of σ based on $y[0], y[1], \ldots, y[N-1]$. That is, taking $\rho = \rho^\star$ in (22) leads to an asymptotically optimal unbiased σ estimation.

Note that $f[n]$ used for calculating $\rho^\star$ is unavailable. Thus, we replace $f[n]$ with its noisy version $y[n]$, attaining the following practically usable coefficient:

$$\rho^\star \approxeq \frac{\sum_{n=\hat{n}}^{N-1} y[n]^2(\mu-\delta_x n)^4}{\sum_{n=0}^{N-1} y[n]^2(\mu-\delta_x n)^4}. \tag{25}$$

If we define A^2/σ_ξ^2 as the estimation SNR, then the proposed σ estimation can be approached as the estimation SNR increases.

Improving the estimation performance of A and μ

So far, we have focused on introducing the novel estimation of σ. From (18), we can see that the proposed σ estimation requires the estimations of A and μ, i.e., $\hat{A}$ and $\hat{n}\delta_x$ therein. To ensure a clear logic flow, we used the naive way of estimating these two parameters, as described in (12). Here, we illustrate some simple yet more accurate methods for estimating A and μ.

For μ estimation, we introduce a local averaging to reduce the impact of noise. Define a rectangular window function as $W[n] = (1/L)$ for $n = 0, 1, \ldots, L-1$ and $W[n] = 0$ for other ns. The local averaging can be performed by using the window function to filter the sampled Gaussian function, i.e., $y[n]$ given in (4). An improved μ estimation can be achieved by searching for the peak of the filtered Gaussian function and then constructing using the peak index. This is expressed as

$$\hat{\mu} = \left(\hat{n} + \left\lfloor \frac{L}{2} \right\rfloor\right)\delta_x, \text{ s.t. } \hat{n} : \operatorname{argmax}_n \sum_{l=0}^{L-1} W[l]\, y[n+l]. \tag{26}$$

The offset $\lfloor L/2 \rfloor$ is added because, in theory, if the sum of continuous L samples is maximum, those samples would be centered around the peak of a Gaussian function. Based on (26), we also obtain an estimate of A, i.e.,

$$\hat{A} = y\left[\hat{n} + \left\lfloor \frac{L}{2} \right\rfloor\right]. \tag{27}$$

Use these two estimates obtained in the proposed σ estimators, as given in (18). Then, combine the two estimates, as done in (22), with the optimal combining coefficient given in (25). This results in the final σ estimate. Unlike $\hat{\mu}$ and $\hat{\sigma}$ obtained using multiple samples, $\hat{A}$ given in (27) is based on a single sample and, hence, can suffer from a large estimation error. Noticing

$\left(xe^{-\frac{(n\delta_x-\hat{\mu})^2}{2\hat{\sigma}^2}}-y[n]\right)^2$,

to x. The solution to the
n is an improved A esti-
en by

$$\left(\sum_{n=0}^{N-1} e^{-\frac{(n\delta_x-\hat{\mu})^2}{2\hat{\sigma}^2}} y[n]\right) \Big/ \left(\sum_{n=0}^{N-1}\left(e^{-\frac{(n\delta_x-\hat{\mu})^2}{2\hat{\sigma}^2}}\right)^2\right). \quad (28)$$

narize the proposed Gauss-
ethod in Table 1 under the
M3. As mentioned at the end
e WLS Fitting" section, we
method as an initial-stage
the second stage, we per-
rative WLS with the initial
ector, i.e., $\mathbf{w}_0$ in (9), con-
g our initial fitting results.
ge fitting is named *M4* in
underline that, due to the
of the proposed initializa-
converge much faster than
iterative WLS, named *M5*
his will be validated shortly
n results. Also provided in
he σ estimation method re-
e "An Interesting but Not-
h Initialization" section,
1. Moreover, the combina-
nd the iterative WLS is re-
M2 in Table 1.

results

results are presented next
he performance of the five
ting methods summarized
The MATLAB simulation
enerating Figures 2 and 3 can be downloaded from https://www.icloud.com/iclouddrive/005qLeEe1YOghnz4Om24ct76w\#publish_v2. Unless otherwise specified, the parameters summarized in Table 2 are primarily used in our simulations. For the original iterative WLS, we perform 12 iterations so that it can achieve a similar asymptotic performance as M4 in the high-SNR region. (This will be seen shortly in Figure 2.) In contrast, when the initialization from either M1 or (the proposed) M3 is employed, only two iterations are performed for the iterative WLS algorithm. Note that the running time for each method is provided in Table 1, where each time result is averaged over 10^5 independent trials.

Figure 2 plots the MSEs of the five estimators listed in Table 1 against the estimation SNR, as given by A^2/σ_ξ^2. Note that μ and σ are randomly generated for the 10^5 independent trials, conforming to the uniform distributions given in Table 2. As given in Table 2, $x \in [0, 10]$ is set in the simulation. Thus, the settings of μ and σ make the Gaussian function to be fitted in each trial incompletely sampled with a long tail; a noisy version of the function is plotted in Figure 1(a).

From Figure 2(b), we see that M3, which is based on the proposed local averaging in (26), achieves an obviously better μ estimation performance than M1, which is based on the naive method given in (12). From Figure 2(c), we see that M3 substantially outperforms M1. This validates the competency of the proposed σ estimation scheme for the cases with the Gaussian function (to be fitted) incompletely sampled. From Figure 2(a), we see a significant improvement in M3, as compared with M1. This validates the advantage of using all samples for A estimation, as developed in (28).

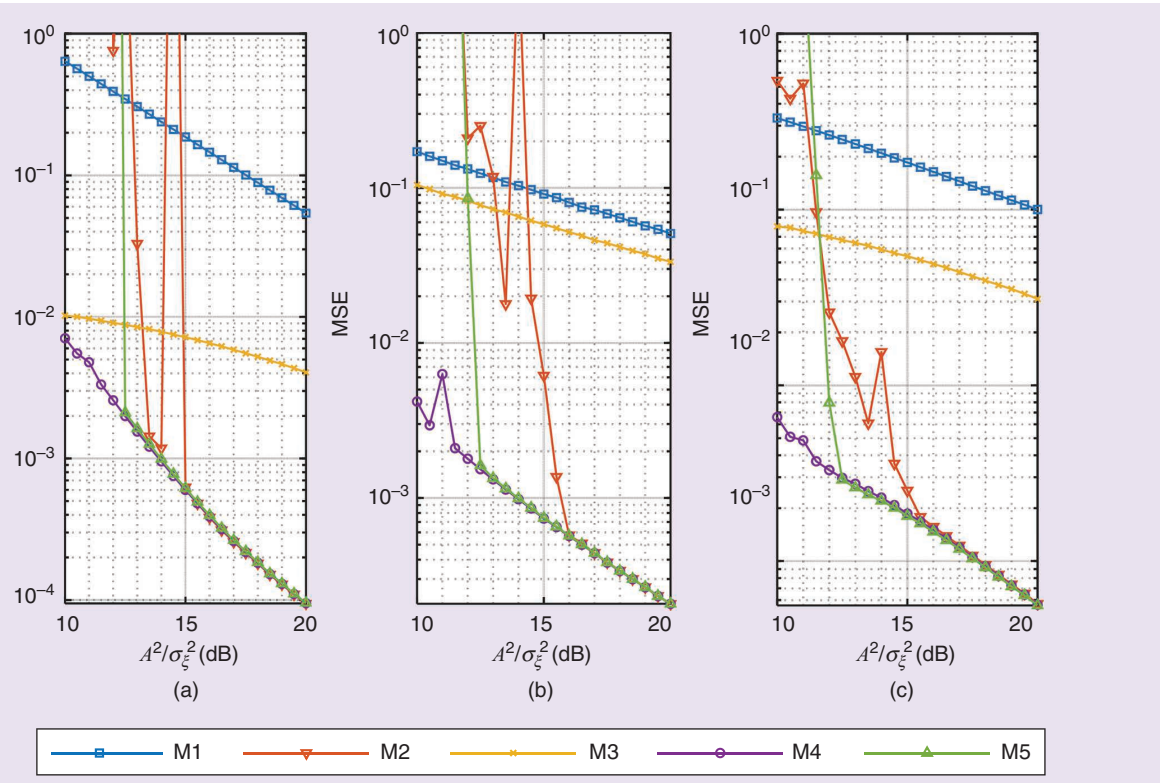
We remark that, as a price paid for performance improvement, the proposed initial fitting requires slightly more computational time than M1; see the last column of Table 1 for comparison. However, it is noteworthy that, thanks to the signal processing tricks introduced in the "Efficient Estimation of σ" section, the proposed σ estimator, as established in (18), only involves simple floating point arithmetic that can be readily handled by modern digital signal processors or field programming gate arrays.

From Figure 2, we further see that the proposed initial fitting results enable the iterative WLS to achieve much better performance for all three parameters than other initializations. The improvement is particularly noticeable in low-SNR regions. Moreover, we underline that M4 based on the proposed initialization only runs two iterations, while the original iterative WLS, i.e., M5, runs 12 iterations. This, on the one hand, illustrates the critical importance of improving the initialization for the iterative WLS, a main motivation of this work. On the other hand, this validates our success in developing a high-quality initialization for the iterative WLS.

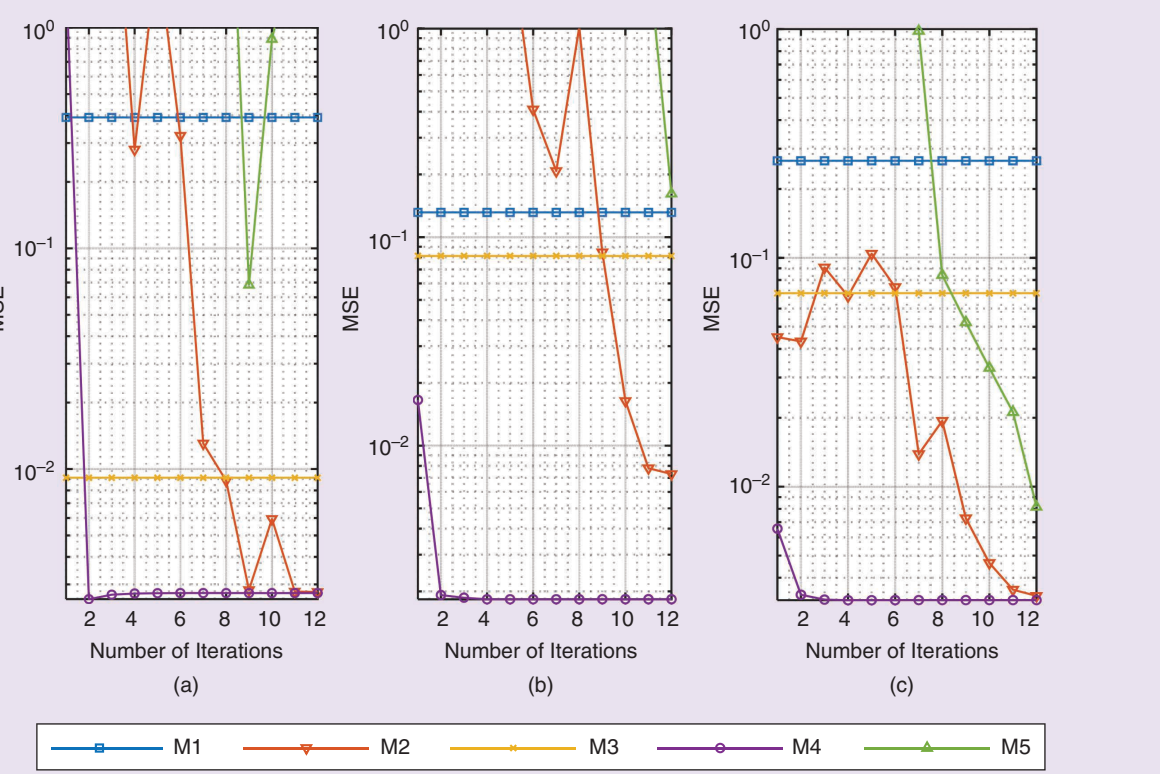
In spite of the random changing of μ and σ over 10^5 independent trials, Figure 2 shows that our proposed initial fitting enables WLS to achieve

mmary of the simulated methods, where the running time of each method is averaged over 10^5 independent trials.

Method	Fitting Steps	Time (μs)
[6]	Estimate $\hat{A}$ and $\hat{n}$ as done in (12), where $\hat{n}$ leads to $\hat{\mu} = \hat{n}\delta_x$; estimate $\hat{\sigma}$ using (11).	76.79
[6] and [5]	Stage 1: Run M1 first, getting initial $\hat{A}$, $\hat{\mu}$, and $\hat{\sigma}$. Stage 2: Perform the iterative WLS based on (9), where $\mathbf{w}_0 = \left[\hat{A}e^{-\frac{(n\delta_x-\hat{n}\delta_x)^2}{2\hat{\sigma}^2}}\Big\|_{n=0,1,\ldots,N-1}\right]^{\mathrm{T}}$.	275
New	Estimate $\hat{\mu}$ based on (26); estimate $\hat{A}$ using (27); perform the estimators in (18), attaining σ_α and σ_β; combine the two estimates in the linear manner depicted in (22), where the optimal ρ, as approximately calculated in (25), is used as the combination coefficient; and refine $\hat{A}$ as done in (28).	304.61
New and [5]	Stage 1: Run M3 first, getting initial $\hat{A}$, $\hat{\mu}$, and $\hat{\sigma}$.	502.81



MSEs of fitting results versus the SNR in the sampled Gaussian function with $A = 1$, where μ and σ are randomly generated based on tributions given in Table 2, for (a) $\hat{A}$, (b) $\hat{\mu}$, and (c) $\hat{\sigma}$. Note that σ_ξ^2 denotes the power of the noise term $\xi[n]$ given in (4). The MSE is 10^5 trials, each with independently generated and normally distributed noise.



better and more stable per-
an prior arts. This suggests
e successfully relieved the
of the iterative WLS on the
f the tail region of a Gauss-
. In contrast, as illustrated
tive WLS Fitting" and "An
ut Not-Good-Enough Ini-
sections, the performance
M1 can be subject to how
Gaussian function is sam-
hich is based on M1, also
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shows the MSEs of the five
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iterations of the iterative
rmed in the second stage
nd M5. For all three pa-
e can see that the proposed
(M3) nontrivially outper-
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, we develop a high-quality
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s achieved by a few signal
icks, as summarized here:
duce a simple local averag-
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n estimating the peak loca-
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Table 2. The simulation parameters.

Variable	Description	Value
A	A parameter of the Gaussian function determining its maximum amplitude [see (1)]	1
μ	A parameter of the Gaussian function indicating its peak location [see (1)]	$\mathcal{U}$ [8,9]
σ	A parameter of the Gaussian function indicating its width of the principal region [see Figure 1(c)]	$\mathcal{U}$ [1,1.3]
x	Function variable	[0,10]
δ_x	Sampling interval of x [see (4)]	0.01
k	Intermediate variable used in the proposed estimators given in (18)	0.1 : 0.01 : 10
L	Windows size for estimating μ as done in (26)	3
	Number of iterations for M5	12
	Number of iterations in stage 2 of M2/M4	2
σ_ξ^2	Power of the additive noise $\xi[n]$ given in (4)	−10 : 0.5 : 20 dB

$\mathcal{U}[a,b]$ denotes the uniform distribution in $[a,b]$.

- We also improve the estimation of the peak amplitude of a Gaussian function by minimizing the MSE of the initial Gaussian fitting.

Corroborated by simulation results, the proposed initialization can substantially improve the accuracy of the iterative WLS-based linear Gaussian fitting, even in challenging scenarios with strong noises and the incompletely sampled Gaussian function with a long tail. Notably, the performance improvement is also accompanied by improved fitting efficiency.

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Authors

Kai Wu (kai.wu@uts.edu.au) received his Ph.D. degree from Xidian University, China, in 2019 and his Ph.D. degree from the University of Technology Sydney (UTS), Australia. His Xidian Ph.D. won the Excellent Ph.D. Thesis Award 2019 from the Chinese Institute of Electronic Engineering. His UTS Ph.D. was awarded The Chancellor's List 2020. His research interests include array signal processing and its applications in radar and communications. He is a Member of IEEE.

J. Andrew Zhang (andrew.zhang@uts.edu.au) received his Ph.D. degree from the Australian National University in 2004. He is an associate professor in the School of Electrical and Data Engineering, University of Technology Sydney, Sydney, NSW 2007, Australia. He was a researcher with Data61, Commonwealth Scientific and Industrial Research Organization (CSIRO), Australia, from 2010 to 2016; Networked Systems, National ICT Australia, Australia, from 2004 to 2010; and ZTE Corp., Nanjing, China, from 1999 to 2001. He has published more than 200 papers in leading international journals and conference proceedings and has won five best paper awards. He received the CSIRO Chairman's Medal and the Australian Engineering Innovation Award in 2012 for exceptional research achievements in multigigabit wireless communications. His research interests include the area of signal processing for wireless communications and sensing. He is a Senior

iversity, Xi'an, China, in
a distinguished professor
director of the Global Big
nologies Centre at the
of Technology Sydney,
W 2007, Australia, and the
rector of the New South
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has won a number of pres-
rds, including Australian
Excellence Awards (2007
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and 2012). He was named
ost influential engineers in
2014 and 2015 and one of

the top researchers in Australia in 2020 and 2021. His research interests include antennas, millimeter-wave and terahertz communications and sensing systems, and big data technologies. He is a Fellow of IEEE.

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HE EDITOR *(continued from page 3)*

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If you have any publication ideas for *SPM*, I encourage you to contact the area editors or me (see the editorial board in the *SPM* web pages) to discuss your idea. Remember that the articles in *SPM* are not suited for the publishing of either new results or surveys. They are tutorial-like articles that must be comprehensive for a wide audience and accompanied by a relevant selection of both figures and references as Robert Heath, the previous *SPM* editor-in-chief explained very clearly in [4]. Concerning the "Lecture Notes" and Tips & Tricks" columns, I also think that sharing data and codes would be an actual added value to these articles, and I encourage authors to use Code Ocean facilities (https://innovate.ieee.org/ieee-code-ocean/) for this purpose.

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ber of academic publishing
He joined IEEE initially as
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he Intellectual Property
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ing at Institut National Polytechnique de Grenoble, France. He has been a professor since 1989 and an emeritus professor since 2019 at Université Grenoble Alpes, Grenoble 38402 France. Since 1979,

neering, speech processing, hyperspectral imaging, and chemical engineering. Since 2019, he has been a scientific advisor for scientific integrity for the French National Center of Scientific